Computer Science Practical List 4

1. Write a Python code with the following functions performing mentioned operations on a binary file "CANDIDATE.DAT" containing lists with [<Candidate no of type int>,<Cname of type string>,<Score of type float>]
   a. Enrol(), to add details of new CANDIDATEs from the user and save in "CANDIDATE.DAT".
   b. GetPass(), to read content from "CANDIDATE.DAT" and display details of those candidates whose Score is >=50.
   c. ShowAll(), to display the entire content of "CANDIDATE.DAT".
   d. AverageScore(), to find and return the average of Score from "CANDIDATE.DAT".

   Also, write menu driven options to call the above functions using a loop.

2. Write a Python code with the following functions performing mentioned operations on a binary file "ACCOUNTS.DAT" containing lists with [<Ano of type int>,<Name of type string>,<Balance of type float>]
   a. Register(), to add details of new account holders as entered by the user in the binary file named ACCOUNTS.DAT.
   b. Transact( ), to perform a transaction i.e. Deposit or Withdraw an Amount in an Account Holders Balance whose Acno is entered by the user from ACCOUNTS.DAT.
      Note: an Account Holder must always have a minimum Balance of Rs 500 in his/her Account.
   c. DisplayAll( ) to display the details of all Account Holders from ACCOUNTS.DAT
   d. BankBalance() to find and return the sum of Balance of all the account holders from ACCOUNTS.DAT

   Also, write a menu driven code to call the above functions using a loop.

3. Write a Python code with the following functions performing mentioned operations on a binary file "EMPLOYEES.DAT" containing lists in the format

   [<Empno>,<Ename>,<Department>, <Salary>]
   a. Register(), to add details of new EMPLOYEESs from the user and save in "EMPLOYEES.DAT".
   b. GetAccounts(), to read content from "EMPLOYEES.DAT" and display details of those Employees whose Department is 'ACCOUNTS'
   c. ShowAll(), to display the entire content of "EMPLOYEES.DAT"
   d. IncreaseSal(), to increase the salary of Employees whose Salary is less than 50000 by 10000 and rewrite the edited content for such employees in the same file.

   Also, write menu driven options to call the above functions in the following order a,b,c,d,c.

4. Write a Python code with the following functions performing mentioned operations on a binary file "ITEMS.DAT" containing lists in the format

   [<Itemno>,<Iname>,<Quantity>, <Price>]
   a. AddItem(), to add details of new Items from the user and save in "ITEMS.DAT".
   b. ShowItems(File), to display the entire content of the File whose name is passed to the parameter.
   c. CreateDemand(), to read the contents of the file ITEMS.DAT and create a new file named LOWSTOCK.DAT which should store the details of all such Items whose quantity in Items.DAT is less than 10.

   Also, write menu driven options to call the above functions in the following order a,b with parameter as 'ITEMS.DAT' ,c,b with parameter as 'LOWSTOCK.DAT'.

5. Write a Python code with the following functions performing mentioned operations on a binary file "BOOKS.DAT" containing lists in the format

   [<Bno>,<Bname>,<Author>, <Price>, <PurchaseDate>]
   a. AddBook(), to add details of new Items from the user and save in "BOOKS.DAT".
   b. ShowBooks(), to display the entire content of the File BOOKS.DAT.
   c. DeleteOld(), to read the contents of the file BOOKS.DAT and delete all such records of the Books whose PurchaseDate is before '1-Jan-2025'.

   Also, write menu driven options to call the above functions in the following order a,b,c,b.

IMPORTANT: REFER TO MORE QUESTIONS GIVEN IN ASSIGNMENT BOOKLET

General Instructions:

i. Type the solution of all the problems using **Python** Language on **Python IDLE** or **Colab**

ii. Type in the following on top of each of your program as comment lines

```
# ------------------------------------------------------------#
# List-Program No      : L4-P1
# Developed By         : Reema Jones
# Date                 : 27-August-2025
# ------------------------------------------------------------#
```

iii. On successful execution with meaningful data, **copy** and **paste** the sample output at the bottom of the program as comment lines.

iv. **"Turn in"** the softcopy of the programs as a single document in response to this assignment submission with each program on separate pages. Once verified by the teacher, take out a hardcopy of **each program on separate pages** from the printer and get them signed by the respective computer teacher. Add all the printouts in the practical file and get the Index entry also signed.

v. Recommended Font, "**Courier New**", Style "**Bold**" Size "**10**" for all programs with single line spacing. Default indentation **2 Character** only.

```
# ------------------------------------------------------------#
# List-Program No : L4-P1
# Developed By    : Puja Dey
# Date            : 31-August-2025
# ------------------------------------------------------------#

import pickle

def Enrol():
    with open("CANDIDATE.DAT", "ab") as f:
        while True:
            num = int(input("Enter Candidate No: "))
            cname = input("Enter Candidate Name: ")
            score = float(input("Enter Score: "))
            pickle.dump([num, cname, score], f)
            ch = input("More records (Y/N)?: ")
            if ch.upper() == "N":
                break

def ShowAll():
    try:
        with open("CANDIDATE.DAT", "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    print(data)
                except EOFError:
                    break
    except FileNotFoundError:
        print("No File!")

def GetPass():
    try:
        with open("CANDIDATE.DAT", "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    if data[2] >= 50:
                        print(data)
                except EOFError:
                    break
    except FileNotFoundError:
        print("No File!")
```

```
def AverageScore():
    try:
        total, count = 0, 0
        with open("CANDIDATE.DAT", "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    total += data[2]
                    count += 1
                except EOFError:
                    break
        if count > 0:
            print("Average Score:", total / count)
        else:
            print("No records!")
    except FileNotFoundError:
        print("No File!")

while True:
    print("\n1. Enrol\n2. ShowAll\n3. GetPass\n4. AverageScore\n5. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        Enrol()
    elif ch == 2:
        ShowAll()
    elif ch == 3:
        GetPass()
    elif ch == 4:
        AverageScore()
    elif ch == 5:
        break
```

```
1. Enrol
2. ShowAll
3. GetPass
4. AverageScore
5. Exit
Enter choice: 2
[1, 'Alpha', 44.0]
[2, 'Beta', 66.0]
[3, 'Gamma', 88.0]

1. Enrol
2. ShowAll
3. GetPass
4. AverageScore
5. Exit
```

```
Enter choice: 1
Enter Candidate No: 4
Enter Candidate Name: Delta
Enter Score: 99
More records (Y/N)?: Y
Enter Candidate No: 55
Enter Candidate Name: Epsilon
Enter Score: 12
More records (Y/N)?: N

1. Enrol
2. ShowAll
3. GetPass
4. AverageScore
5. Exit
Enter choice: 2
[1, 'Alpha', 44.0]
[2, 'Beta', 66.0]
[3, 'Gamma', 88.0]
[4, 'Delta', 99.0]
[55, 'Epsilon', 12.0]

1. Enrol
2. ShowAll
3. GetPass
4. AverageScore
5. Exit
Enter choice: 3
[2, 'Beta', 66.0]
[3, 'Gamma', 88.0]
[4, 'Delta', 99.0]

1. Enrol
2. ShowAll
3. GetPass
4. AverageScore
5. Exit
Enter choice: 4
Average Score: 61.8

1. Enrol
2. ShowAll
3. GetPass
4. AverageScore
5. Exit
Enter choice: 5
```

```
# ------------------------------------------------------------#
# List-Program No : L4-P2
# Developed By    : Puja Dey
# Date            : 31-August-2025
# ------------------------------------------------------------#

import pickle
def Register():
    with open("ACCOUNTS.DAT", "ab") as f:
        while True:
            ano = int(input("Enter Account No: "))
            name = input("Enter Name: ")
            balance = float(input("Enter Balance: "))
            pickle.dump([ano, name, balance], f)
            ch = input("More records (Y/N)?: ")
            if ch.upper() == "N":
                break

def DisplayAll():
    try:
        with open("ACCOUNTS.DAT", "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    print(data)
                except EOFError:
                    break
    except FileNotFoundError:
        print("No File!")

def Transact():
    try:
        records = []
        acno = int(input("Enter Account No for transaction: "))
        with open("ACCOUNTS.DAT", "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    if data[0] == acno:
                        print("Current Balance:", data[2])
                        t = input("Deposit or Withdraw (D/W)?: ").upper()
                        amt = float(input("Enter Amount: "))
                        if t == "D":
```

```
                            data[2] += amt
                        elif t == "W":
                            if data[2] - amt >= 500:
                                data[2] -= amt
                            else:
                                        print("Insufficient balance! Minimum 500 must
      remain.")
                        print("Updated Balance:", data[2])
                    records.append(data)
                except EOFError:
                    break
        with open("ACCOUNTS.DAT", "wb") as f:
            for r in records:
                pickle.dump(r, f)
    except FileNotFoundError:
        print("No File!")

def BankBalance():
    try:
        total = 0
        with open("ACCOUNTS.DAT", "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    total += data[2]
                except EOFError:
                    break
        print("Total Bank Balance:", total)
    except FileNotFoundError:
        print("No File!")

while True:
    print("\n1. Register\n2. DisplayAll\n3. Transact\n4. BankBalance\n5. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        Register()
    elif ch == 2:
        DisplayAll()
    elif ch == 3:
        Transact()
    elif ch == 4:
        BankBalance()
    elif ch == 5:
        break
```

```
Enter Account No: 2
Enter Name: Beta
Enter Balance: 2000
More records (Y/N)?: Y
Enter Account No: 3
Enter Name: Gamma
Enter Balance: 800
More records (Y/N)?: N

1. Register
2. DisplayAll
3. Transact
4. BankBalance
5. Exit
Enter choice: 2
[1, 'Alpha', 5000.0]
[2, 'Beta', 2000.0]
[3, 'Gamma', 800.0]

1. Register
2. DisplayAll
3. Transact
4. BankBalance
5. Exit
Enter choice: 3
Enter Account No for transaction: 2
Current Balance: 2000.0
Deposit or Withdraw (D/W)?: D
Enter Amount: 1000
Updated Balance: 3000.0

1. Register
2. DisplayAll
3. Transact
4. BankBalance
5. Exit
Enter choice: 3
Enter Account No for transaction: 3
Current Balance: 800.0
Deposit or Withdraw (D/W)?: W
Enter Amount: 500
Insufficient balance! Minimum 500 must remain.
Updated Balance: 800.0

1. Register
2. DisplayAll
3. Transact
4. BankBalance
5. Exit
Enter choice: 4
Total Bank Balance: 8800.0

1. Register
2. DisplayAll
3. Transact
4. BankBalance
5. Exit
Enter choice: 5
```

```
# ------------------------------------------------------------#
# List-Program No : L4-P3
# Developed By    : Puja Dey
# Date            : 31-August-2025
# ------------------------------------------------------------#

import pickle

def Register():
    with open("EMPLOYEES.DAT", "ab") as f:
        while True:
            empno = int(input("Enter Employee No: "))
            ename = input("Enter Employee Name: ")
            dept = input("Enter Department: ")
            sal = float(input("Enter Salary: "))
            pickle.dump([empno, ename, dept, sal], f)
            ch = input("More records (Y/N)?: ")
            if ch.upper() == "N":
                break

def ShowAll():
    try:
        with open("EMPLOYEES.DAT", "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    print(data)
                except EOFError:
                    break
    except FileNotFoundError:
        print("No File!")

def GetAccounts():
    try:
        with open("EMPLOYEES.DAT", "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    if data[2].upper() == "ACCOUNTS":
                        print(data)
                except EOFError:
                    break
    except FileNotFoundError:
```

```
        print("No File!")

def IncreaseSal():
    try:
        records = []
        with open("EMPLOYEES.DAT", "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    if data[3] < 50000:
                        data[3] += 10000
                    records.append(data)
                except EOFError:
                    break
        with open("EMPLOYEES.DAT", "wb") as f:
            for r in records:
                pickle.dump(r, f)
    except FileNotFoundError:
        print("No File!")

while True:
    print("\n1. Register\n2. GetAccounts\n3. ShowAll\n4. IncreaseSal\n5. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        Register()
    elif ch == 2:
        GetAccounts()
    elif ch == 3:
        ShowAll()
    elif ch == 4:
        IncreaseSal()
    elif ch == 5:
        break
```

```
Enter Employee No: 101
Enter Employee Name: Alpha
Enter Department: Accounts
Enter Salary: 45000
More records (Y/N)?: Y
Enter Employee No: 102
Enter Employee Name: Beta
Enter Department: HR
Enter Salary: 60000
More records (Y/N)?: Y
Enter Employee No: 103
Enter Employee Name: Gamma
Enter Department: Accounts
Enter Salary: 30000
```

```
More records (Y/N)?: N

1. Register
2. GetAccounts
3. ShowAll
4. IncreaseSal
5. Exit
Enter choice: 2
[101, 'Alpha', 'Accounts', 45000.0]
[103, 'Gamma', 'Accounts', 30000.0]

1. Register
2. GetAccounts
3. ShowAll
4. IncreaseSal
5. Exit
Enter choice: 3
[101, 'Alpha', 'Accounts', 45000.0]
[102, 'Beta', 'HR', 60000.0]
[103, 'Gamma', 'Accounts', 30000.0]

1. Register
2. GetAccounts
3. ShowAll
4. IncreaseSal
5. Exit
Enter choice: 4

1. Register
2. GetAccounts
3. ShowAll
4. IncreaseSal
5. Exit
Enter choice: 3
[101, 'Alpha', 'Accounts', 55000.0]
[102, 'Beta', 'HR', 60000.0]
[103, 'Gamma', 'Accounts', 40000.0]

1. Register
2. GetAccounts
3. ShowAll
4. IncreaseSal
5. Exit
Enter choice: 5
```

```
# ------------------------------------------------------------#
# List-Program No : L4-P4
# Developed By    : Puja Dey
# Date            : 31-August-2025
# ------------------------------------------------------------#

import pickle

def AddItem():
    with open("ITEMS.DAT", "ab") as f:
        while True:
            ino = int(input("Enter Item No: "))
            iname = input("Enter Item Name: ")
            qty = int(input("Enter Quantity: "))
            price = float(input("Enter Price: "))
            pickle.dump([ino, iname, qty, price], f)
            ch = input("More records (Y/N)?: ")
            if ch.upper() == "N":
                break

def ShowItems(File):
    try:
        with open(File, "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    print(data)
                except EOFError:
                    break
    except FileNotFoundError:
        print("No File!")

def CreateDemand():
    try:
        with open("ITEMS.DAT", "rb") as f, open("LOWSTOCK.DAT", "wb") as g:
            while True:
                try:
                    data = pickle.load(f)
                    if data[2] < 10:
                        pickle.dump(data, g)
                except EOFError:
                    break
    except FileNotFoundError:
```

```
        print("No File!")

while True:
            print("\n1.   AddItem\n2.   ShowItems(ITEMS.DAT)\n3.   CreateDemand\n4.
ShowItems(LOWSTOCK.DAT)\n5. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        AddItem()
    elif ch == 2:
        ShowItems("ITEMS.DAT")
    elif ch == 3:
        CreateDemand()
    elif ch == 4:
        ShowItems("LOWSTOCK.DAT")
    elif ch == 5:
        break
```

```
1. AddItem
2. ShowItems(ITEMS.DAT)
3. CreateDemand
4. ShowItems(LOWSTOCK.DAT)
5. Exit
Enter choice: 3
No File!

1. AddItem
2. ShowItems(ITEMS.DAT)
3. CreateDemand
4. ShowItems(LOWSTOCK.DAT)
5. Exit
Enter choice: 1
Enter Item No: 101
Enter Item Name: PEN
Enter Quantity: 50
Enter Price: 10
More records (Y/N)?: Y
Enter Item No: 102
Enter Item Name: NOTEBOOK
Enter Quantity: 5
Enter Price: 50
More records (Y/N)?: Y
Enter Item No: 103
Enter Item Name: ERASER
Enter Quantity: 8
Enter Price: 5
More records (Y/N)?: N

1. AddItem
2. ShowItems(ITEMS.DAT)
3. CreateDemand
4. ShowItems(LOWSTOCK.DAT)
5. Exit
Enter choice: 2
[101, 'PEN', 50, 10.0]
```

```
[102, 'NOTEBOOK', 5, 50.0]
[103, 'ERASER', 8, 5.0]

1. AddItem
2. ShowItems(ITEMS.DAT)
3. CreateDemand
4. ShowItems(LOWSTOCK.DAT)
5. Exit
Enter choice: 3

1. AddItem
2. ShowItems(ITEMS.DAT)
3. CreateDemand
4. ShowItems(LOWSTOCK.DAT)
5. Exit
Enter choice: 4
[102, 'NOTEBOOK', 5, 50.0]
[103, 'ERASER', 8, 5.0]

1. AddItem
2. ShowItems(ITEMS.DAT)
3. CreateDemand
4. ShowItems(LOWSTOCK.DAT)
5. Exit
Enter choice: 5
```

```
# ------------------------------------------------------------#
# List-Program No : L4-P5
# Developed By    : Puja Dey
# Date            : 31-August-2025
# ------------------------------------------------------------#

import pickle
from datetime import datetime

def AddBook():
    with open("BOOKS.DAT", "ab") as f:
        while True:
            bno = int(input("Enter Book No: "))
            bname = input("Enter Book Name: ")
            author = input("Enter Author: ")
            price = float(input("Enter Price: "))
            pdate = input("Enter Purchase Date (DD-Mon-YYYY): ")
            pickle.dump([bno, bname, author, price, pdate], f)
            ch = input("More records (Y/N)?: ")
            if ch.upper() == "N":
                break

def ShowBooks():
    try:
        with open("BOOKS.DAT", "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    print(data)
                except EOFError:
                    break
    except FileNotFoundError:
        print("No File!")

def DeleteOld():
    try:
        records = []
        cutoff = datetime.strptime("01-Jan-2025", "%d-%b-%Y")
        with open("BOOKS.DAT", "rb") as f:
            while True:
                try:
                    data = pickle.load(f)
                    pdate = datetime.strptime(data[4], "%d-%b-%Y")
                    if pdate >= cutoff:
```

```
                        records.append(data)
                except EOFError:
                    break
        with open("BOOKS.DAT", "wb") as f:
            for r in records:
                pickle.dump(r, f)
    except FileNotFoundError:
        print("No File!")

while True:
    print("\n1. AddBook\n2. ShowBooks\n3. DeleteOld\n4. ShowBooks\n5. Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        AddBook()
    elif ch == 2:
        ShowBooks()
    elif ch == 3:
        DeleteOld()
    elif ch == 4:
        ShowBooks()
    elif ch == 5:
        break
```

```
1. AddBook
2. ShowBooks
3. DeleteOld
4. ShowBooks
5. Exit
Enter choice: 2
No File!

1. AddBook
2. ShowBooks
3. DeleteOld
4. ShowBooks
5. Exit
Enter choice: 1
Enter Book No: 101
Enter Book Name: IRODOV
Enter Author: IE IRODOV
Enter Price: 450
Enter Purchase Date (DD-Mon-YYYY): 15-Dec-2024
More records (Y/N)?: Y
Enter Book No: 102
Enter Book Name: KROTOV
Enter Author: SS KROTOV
Enter Price: 650
Enter Purchase Date (DD-Mon-YYYY): 10-Jan-2025
More records (Y/N)?: Y
Enter Book No: 103
Enter Book Name: PATHFINDER
Enter Author: PEARSON
```

```
Enter Price: 550
Enter Purchase Date (DD-Mon-YYYY): 20-Feb-2025
More records (Y/N)?: N

1. AddBook
2. ShowBooks
3. DeleteOld
4. ShowBooks
5. Exit
Enter choice: 2
[101, 'IRODOV', 'IE IRODOV', 450.0, '15-Dec-2024']
[102, 'KROTOV', 'SS KROTOV', 650.0, '10-Jan-2025']
[103, 'PATHFINDER', 'PEARSON', 550.0, '20-Feb-2025']

1. AddBook
2. ShowBooks
3. DeleteOld
4. ShowBooks
5. Exit
Enter choice: 3

1. AddBook
2. ShowBooks
3. DeleteOld
4. ShowBooks
5. Exit
Enter choice: 4
[102, 'KROTOV', 'SS KROTOV', 650.0, '10-Jan-2025']
[103, 'PATHFINDER', 'PEARSON', 550.0, '20-Feb-2025']

1. AddBook
2. ShowBooks
3. DeleteOld
4. ShowBooks
5. Exit
Enter choice: 5
```